

Python Data Structure Questions with Solutions (Updated)

INT & FLOAT

1. Take two numbers as input and print their difference.

Solution:

```
a = int(input()); b = int(input()); print(a - b)
```

2. Multiply a float with an integer and print the result.

Solution:

```
a = float(input()); b = int(input()); print(a * b)
```

3. Divide a number by 2 and show the remainder.

Solution:

```
n = int(input()); print(n % 2)
```

4. Check if a given number is divisible by 7.

Solution:

```
n = int(input()); print(n % 7 == 0)
```

5. Round a float number to 1 decimal place.

Solution:

```
f = float(input()); print(round(f, 1))
```

6. Convert an integer to a float and print both types.

Solution:

```
i = int(input()); f = float(i); print(i, f)
```

7. Take a float input and print its integer part only.

Solution:

```
f = float(input()); print(int(f))
```

8. Use the modulus operator to get the last digit of a number.

Solution:

```
n = int(input()); print(n % 10)
```

9. Add 10 to an integer and then subtract 5 from it. Print the result.

Solution:

```
n = int(input()); print((n + 10) - 5)
```

10. Calculate simple interest given P, R, T (No loop needed).

Solution:

```
P = float(input()); R = float(input()); T = float(input()); print((P * R * T) / 100)
```

Python Data Structure Questions with Solutions (Updated)

STR (String)

11. Take a name as input and print its first and last characters.

Solution:

```
s = input(); print(s[0], s[-1])
```

12. Convert a string to uppercase.

Solution:

```
s = input(); print(s.upper())
```

13. Convert a string to lowercase.

Solution:

```
s = input(); print(s.lower())
```

14. Find the position of the first occurrence of 'a' in a string.

Solution:

```
s = input(); print(s.find('a'))
```

15. Replace all 'e' characters in a string with '@'.

Solution:

```
s = input(); print(s.replace('e', '@'))
```

16. Concatenate two strings with a space in between.

Solution:

```
s1 = input(); s2 = input(); print(s1 + ' ' + s2)
```

17. Print the length of a given string.

Solution:

```
s = input(); print(len(s))
```

18. Slice the first 3 characters of a string.

Solution:

```
s = input(); print(s[:3])
```

19. Check if a string contains the word 'hello'.

Solution:

```
s = input(); print('hello' in s)
```

20. Count how many times the letter 'i' appears in a string.

Solution:

```
s = input(); print(s.count('i'))
```

Python Data Structure Questions with Solutions (Updated)

LIST

21. Create a list of 3 items and print it.

Solution:

```
lst = ['a', 'b', 'c']; print(lst)
```

22. Access the second element of a list and print it.

Solution:

```
lst = ['x', 'y', 'z']; print(lst[1])
```

23. Replace the last item in a list with a new value.

Solution:

```
lst = [1, 2, 3]; lst[-1] = 10; print(lst)
```

24. Append an item to the list and print it.

Solution:

```
lst = [1, 2]; lst.append(3); print(lst)
```

25. Insert an item at the second position in a list.

Solution:

```
lst = [1, 3, 4]; lst.insert(1, 2); print(lst)
```

26. Remove a specific item from the list.

Solution:

```
lst = [1, 2, 3]; lst.remove(2); print(lst)
```

27. Print the length of the list.

Solution:

```
lst = [1, 2, 3, 4]; print(len(lst))
```

28. Sort the list in ascending order.

Solution:

```
lst = [4, 2, 1]; lst.sort(); print(lst)
```

29. Reverse the list using slicing.

Solution:

```
lst = [1, 2, 3]; print(lst[::-1])
```

30. Join all elements in a list of strings into a single string separated by commas.

Solution:

```
lst = ['a', 'b', 'c']; print(','.join(lst))
```

Python Data Structure Questions with Solutions (Updated)

TUPLE

31. Create a tuple with 3 integers and print it.

Solution:

```
t = (1, 2, 3); print(t)
```

32. Access the first element of a tuple.

Solution:

```
t = (10, 20, 30); print(t[0])
```

33. Slice the tuple to get the last two elements.

Solution:

```
t = (1, 2, 3, 4); print(t[-2:])
```

34. Use the 'in' keyword to check if a value exists in the tuple.

Solution:

```
t = (5, 6, 7); print(6 in t)
```

35. Get the length of the tuple.

Solution:

```
t = (1, 2, 3); print(len(t))
```

36. Convert the tuple to a list.

Solution:

```
t = (1, 2, 3); print(list(t))
```

37. Count how many times a value appears in a tuple.

Solution:

```
t = (1, 2, 2, 3); print(t.count(2))
```

38. Find the index of a given value in the tuple.

Solution:

```
t = (10, 20, 30); print(t.index(20))
```

39. Concatenate two tuples.

Solution:

```
t1 = (1,); t2 = (2,); print(t1 + t2)
```

40. Check if two tuples are equal.

Solution:

```
t1 = (1, 2); t2 = (1, 2); print(t1 == t2)
```

Python Data Structure Questions with Solutions (Updated)

DICT (Dictionary)

41. Create a dictionary with keys: 'name', 'age', 'city'.

Solution:

```
d = {'name': 'Ana', 'age': 22, 'city': 'Pune'}; print(d)
```

42. Access the value of the key 'name'.

Solution:

```
d = {'name': 'Ana'}; print(d['name'])
```

43. Update the value of 'age' in the dictionary.

Solution:

```
d = {'age': 22}; d['age'] = 23; print(d)
```

44. Add a new key 'email' with any value.

Solution:

```
d = {}; d['email'] = 'a@example.com'; print(d)
```

45. Delete the key 'city' from the dictionary.

Solution:

```
d = {'city': 'Pune'}; del d['city']; print(d)
```

46. Print all the keys of the dictionary.

Solution:

```
d = {'a': 1, 'b': 2}; print(d.keys())
```

47. Print all the values of the dictionary.

Solution:

```
d = {'a': 1, 'b': 2}; print(d.values())
```

48. Check if 'age' exists as a key in the dictionary.

Solution:

```
d = {'age': 25}; print('age' in d)
```

49. Get the value of a key using the get() method.

Solution:

```
d = {'name': 'Ana'}; print(d.get('name'))
```

50. Clear all items from the dictionary.

Solution:

```
d = {'a': 1}; d.clear(); print(d)
```

Python Data Structure Questions with Solutions (Updated)

SET

51. Create a set with 3 numbers and print it.

Solution:

```
s = {1, 2, 3}; print(s)
```

52. Add a new element to the set.

Solution:

```
s = {1, 2}; s.add(3); print(s)
```

53. Remove an element from the set.

Solution:

```
s = {1, 2, 3}; s.remove(2); print(s)
```

54. Check if an element exists in the set.

Solution:

```
s = {1, 2, 3}; print(2 in s)
```

55. Find the union of two sets.

Solution:

```
a = {1, 2}; b = {2, 3}; print(a | b)
```

56. Find the intersection of two sets.

Solution:

```
a = {1, 2}; b = {2, 3}; print(a & b)
```

57. Find the difference of two sets.

Solution:

```
a = {1, 2}; b = {2, 3}; print(a - b)
```

58. Copy a set to a new variable.

Solution:

```
s = {1, 2}; s2 = s.copy(); print(s2)
```

59. Check if a set is a subset of another.

Solution:

```
a = {1, 2}; b = {1, 2, 3}; print(a.issubset(b))
```

60. Clear all elements from a set.

Solution:

```
s = {1, 2}; s.clear(); print(s)
```